REPORT

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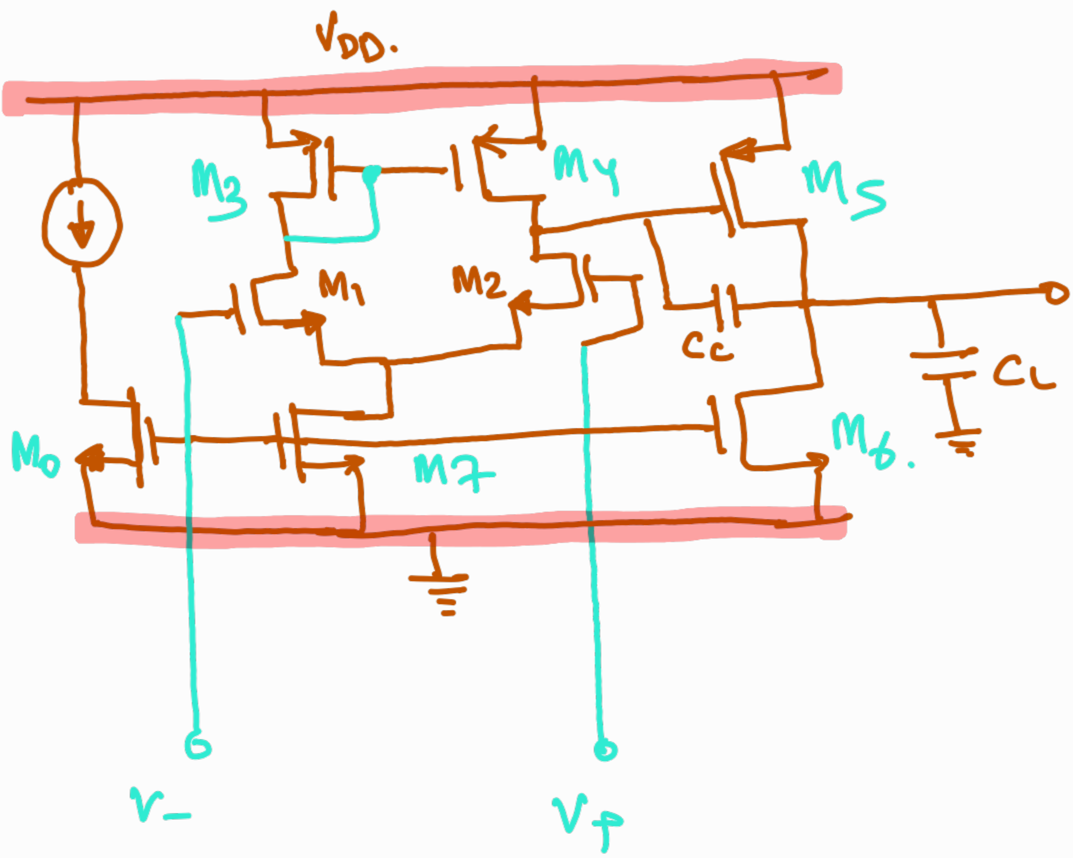
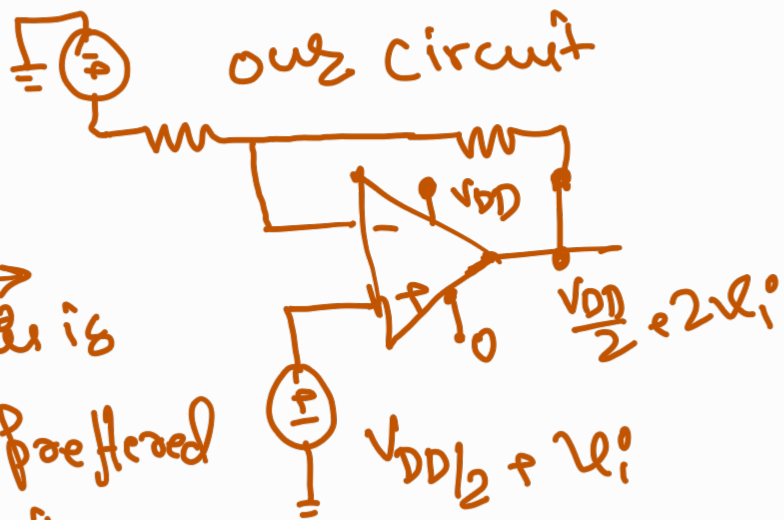
Circuit Diagram



Assumptions and Given informations:-



this is our preffered circuit.

Specifications:-

(i) atleast 40dB (let's take 60dB) of DC gain.

(ii). GBW = 5MHz

(iii) PM $\geq 60^\circ$

(iv) Slew rate $\geq$ 10V/μsec

(v) ICMR(+) = 1.6V ICMR(−) = 0.8V

(vi) lets take $C_L = 10pF$

(vii) V_{dd} = 1.8V.

(viii) $\mu_n C_{ox}$ = 200 $\mu A/V^2$ (initially)

$\mu_p C_{ox}$ = 100 $\mu A/V^2$ (initial)

Calculations:-

Slew rate = $\frac{I_7}{C_c}$

and for $\geq 60^\circ$ phase margin

$$C_c = 0.4 C_L = 4pF.$$

$$I_7 \geq 10 \times 10^{+6} \times 4 \times 10^{-12}$$

$I_7 \geq 40\,\mu A$

M_1 & M_2

$$g_{m1} = GBW \cdot C_c \cdot 2\pi$$

$$g_{m1} = 5 \times 10^6 \times 2\pi \times 4 \times 10^{-12}$$

$$g_{m1} \Rightarrow 40\pi\,\mu s$$

$$g_{m1} = 125\,\mu s$$

$$g_{m1} = \sqrt{I_{D7}\,\mu_n C_{ox} \left(\frac{W}{L}\right)}$$

$$\Rightarrow \left(\frac{W}{L}\right)_1 = \frac{g_{m1}^2}{\underbrace{\mu_n C_{ox}} I_{D7}} = \frac{(125)^2}{230 \times 40}$$

$$= 1.69$$

from the Rough slide we get

$$\left(\frac{W}{L}\right)_2 = 1.69.$$

$$\sqrt{\frac{I_{DS}}{\beta_3}} = V_{DD} - |V_{th3}| + V_{th1} - V_{in})_{max}$$

$$\left(\frac{W}{L}\right)_{3,4} = \frac{I_S}{\mu_p C_{ox} (V_{DD} - |V_{th3}| + V_{th1} - V_{in\,max})^2}$$

$$\left(\frac{W}{L}\right)_{3,4} = \frac{40\,\cancel{\mu A}}{100\,\frac{\cancel{\mu A}}{V^2}(1.8 - 0.37 + 0.37 - 1.6)^2\,\cancel{V^2}}$$

$$\cancel{\left(\frac{W}{L}\right)_{3,4} = 1 \quad 20.40}$$

Since $I_7 \geq 40\mu A$ and $I_{ref} = 4\mu A$

So, 10 times current

So, factor is ≥ 10 Assume

$\left(\frac{W}{L}\right)_{M0} = 2$, then we must

have $\left(\frac{W}{L}\right)_7 \geq 2 \times 10 = 20.$

I took it be 40 for good gain.

for reasonable phase margin ($\sim 60°$)

$$g_{m5} = 2.2\, g_{m4}\left(\frac{C_L}{C_c}\right)!!$$

&

$$g_{m5} >> 10 g_{m1}.$$

$\Rightarrow$ $V_{SG_4} = V_{SG_5}$

$$\left(\frac{W}{L}\right)_5 = \left(\frac{W}{L}\right)_4 \left(\frac{g_{m5}}{g_{m4}}\right)$$

$\left(P_2 = \frac{g_{m5}}{2\pi C_L}\right)$!! (comes purely from the second stage).

$$GBW = 5MHz$$

$$P_2 \geq 2.2 \times 5MHz = 11 MHz$$

$$P_2 \approx \frac{g_{m5}}{2\pi C_L}$$

$$11 \times 10^6 = \frac{g_{m5}}{2\pi \times (10 \times 10^{-6})}$$

$$g_{m5} \geq 691 / \mu s$$

now since overdrive in both are same:-

$(V_{SG} - |V_{tp}|)$ is same both M_4 and M_5;

$$g_{m4} = \sqrt{2 \mu_p C_{ox} (W/L)_4 I_{D4}}$$

$$= \sqrt{2 \times 100\mu \times 20.45 (20)\mu}$$

$$= 286 \mu S$$

$$\Delta V_{o4} = \frac{2 I_{D4}}{g_{m4}} = \frac{2(20 \mu A)}{286 \mu S}$$

$$= 0.14 V$$

so for better phase margin we already got

$$g_{m5} \geq 691 \mu S$$

we must have $\Delta V_5 = \Delta V_4 = 0.14 V$

$$g_{m5} = \frac{2 I_{D5}}{V_{ov5}}$$

$$I_{D5} = \frac{691\mu \times 0.14}{2} = 48.37 \mu A$$

lets assume ($\approx 55 \mu A$)

$$(W/L)_5 = 20.45 \times 2.75 = 56.2375.$$

$W = 25.551\ \mu m$, $L = 0.45/\mu m$

now

M_6 must be able to pull this much current

$\left(\frac{55}{4}\right)$ = multiplier

$\frac{55}{4} \times 1.69 \approx \frac{55}{4} \times 2 = 27.5$

So, $\left(\frac{W}{L}\right)_6 = 27.5$

Now we see what we get from the LTspice:-

(i) dc gain: 64.783 dB (≥ 60 dB).

(ii) Phase margin at 0dB = $65°$

(iii). GBW $\approx$ 4.7 MHz.

(iv) Currents in Each transistor:-

$M_0 = 4\ \mu A$ ← All of these are dc currents.

$M_1 = 27.87\ \mu A$.

$M_2 = 27.74\ \mu A$ (This is little small due to the fact that it single handedly

$M_3 = 27.87\ \mu A$

$M_4 = 27.74\,\mu A$

$M_7 = 55.62\,\mu A.$

$M_5 = 46.8984\,\mu A$

$M_6 = 46.8984\,\mu A$

justifies the saturation of the M_5, M_6 transistor for maximum gain

$\left(\frac{W}{L}\right)_0 = 2$; $\left(\frac{W}{L}\right)_7 = 40$ $\left(\frac{W}{L}\right)_{1,2} = 1.69$

$\left(\frac{W}{L}\right)_{3,4} = 20.45$ $\left(\frac{W}{L}\right)_5 = 51.1$ $\left(\frac{W}{L}\right)_6 = 25\,\mu$

now ; if we calculate the g_m and the r_o

we have $r_o = \dfrac{1}{\lambda I_D}$ &

$$g_m = \sqrt{2\mu_n C_{ox}\left(\frac{W}{L}\right).I_D}$$

Tr No.	r_o (KΩ)	g_m (μS)	ΔV (V)	W/L	I_D (μA)
M_0	2500	60.7	0.132	2.0	4.0
M_1	360	147	0.378	1.69	27.8
M_2	360	147	0.378	1.69	27.8
M_3	360	337.2	0.165	20.45	27.8
M_4	360	337.2	0.165	20.45	27.8
M_5	180	1011.6	0.110	40	46.9
M_6	213	692.3	0.135	51.1	46.9
M_7	213	734.4	0.128	25.5	55.5

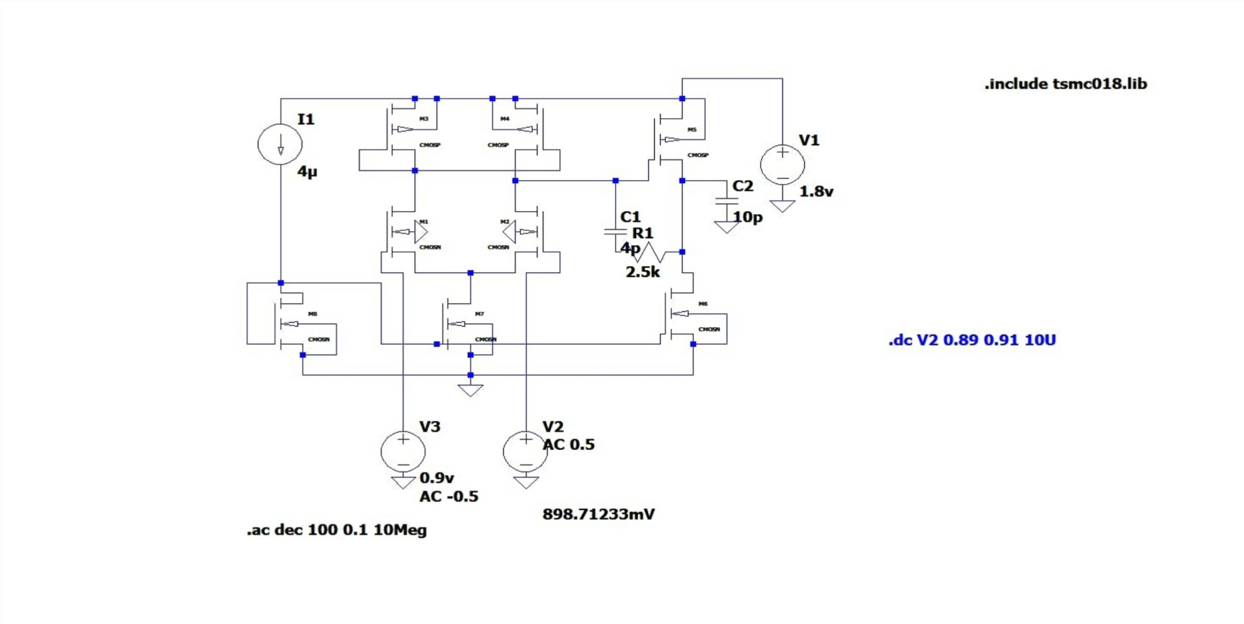
Tr	W	L
M0	1.0 um	0.5 um
M1	0.855 um	0.5 um
M2	0.855 um	0.5 um
M3	10.22 um	0.5 um
M4	10.22 um	0.5 um
M5	25.55 um	0.5 um

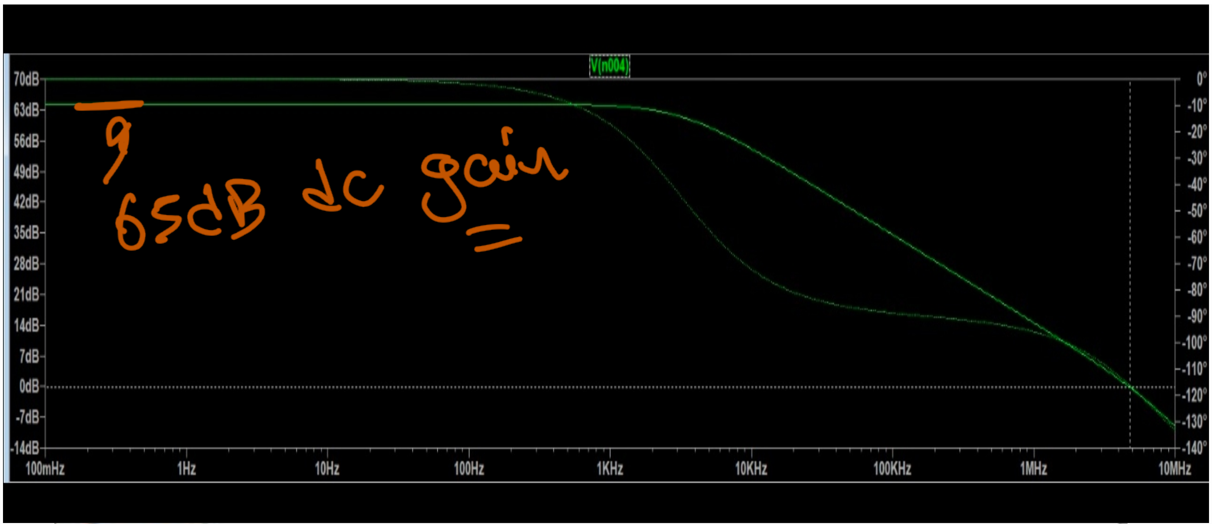
everything in um.

M6	12.5 um	0.5 um
M7	20 um	0.5 um

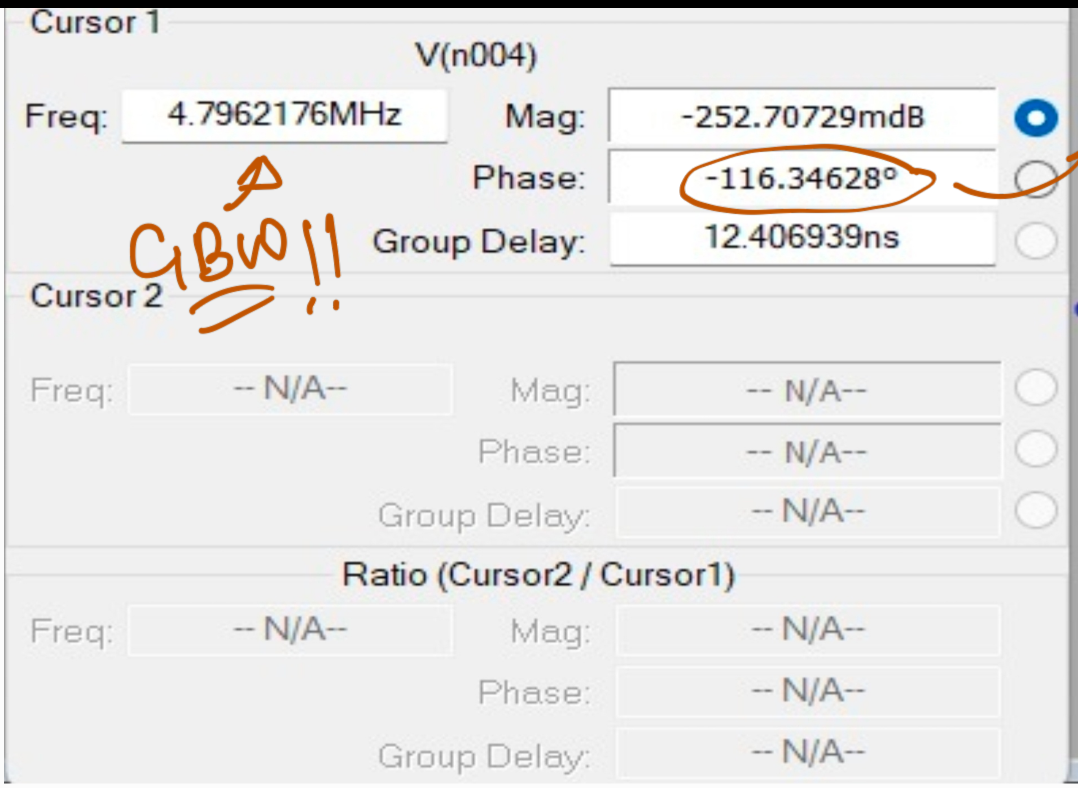
Diagrams and Simulations:-

Final circuit :- open loop characteristics:-





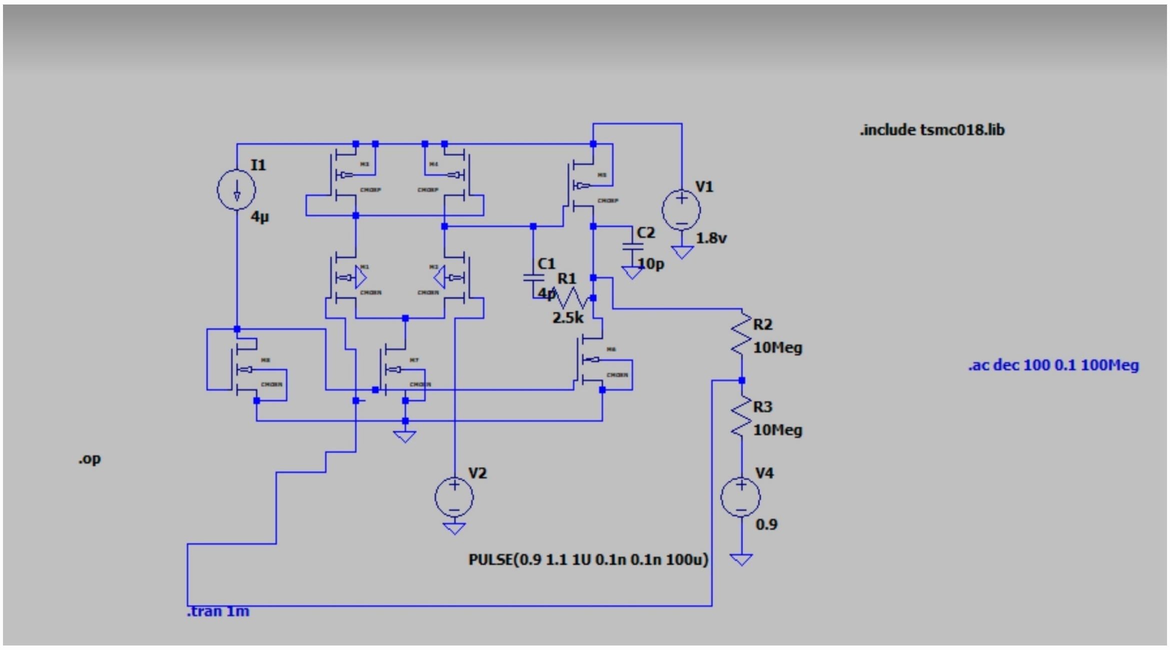
65dB dc gain



GBW!!

≈65° PM

Closed loop :-



Transient response :-

Operating points :-

V(n001):	1.8	voltage
V(n002):	1.15744	voltage
V(n003):	1.19874	voltage
V(n004):	0.902604	voltage
V(n005):	0.9	voltage
V(n006):	0.902604	voltage
V(n007):	0.901302	voltage
V(n008):	0.0849597	voltage
V(n009):	0.557781	voltage
V(n010):	0.9	voltage
I(C1):	1.18453e-24	device_current
I(C2):	9.02604e-24	device_current
I(I1):	4e-06	device_current
I(R1):	0	device_current
I(R2):	1.30188e-10	device_current
I(R3):	1.30188e-10	device_current
I(V1):	-0.000106646	device_current
I(V2):	0	device_current
I(V4):	1.30188e-10	device_current
Ib(M1):	-1.26203e-12	device_current
Ib(M2):	-1.30332e-12	device_current
Ib(M3):	6.52559e-13	device_current
Ib(M4):	6.11263e-13	device_current
Ib(M5):	9.07396e-13	device_current
Ib(M6):	-9.12604e-13	device_current
Ib(M7):	-9.45853e-14	device_current
Ib(M8):	-5.67781e-13	device_current

Ib(M8): -5.67781e-13 device_current

Id(M1):	2.79511e-05	device_current
Id(M2):	2.78263e-05	device_current
Id(M3):	2.79511e-05	device_current
Id(M4):	2.78263e-05	device_current
Id(M5):	4.68687e-05	device_current
Id(M6):	4.68686e-05	device_current
Id(M7):	5.57773e-05	device_current
Id(M8):	4e-06	device_current
Ig(M1):	0	device_current
Ig(M2):	0	device_current
Ig(M3):	-0	device_current
Ig(M4):	-0	device_current
Ig(M5):	-0	device_current
Ig(M6):	0	device_current
Ig(M7):	0	device_current
Ig(M8):	0	device_current
Is(M1):	-2.79511e-05	device_current
Is(M2):	-2.78263e-05	device_current
Is(M3):	-2.79511e-05	device_current
Is(M4):	-2.78263e-05	device_current
Is(M5):	-4.68687e-05	device_current
Is(M6):	-4.68686e-05	device_current
Is(M7):	-5.57773e-05	device_current
Is(M8):	-4e-06	device_current